KAJAL SANWAL

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience building scalable full-stack and embedded systems using React, Next.js, Node.js, and PostgreSQL. Strong foundation in Data Structures & Algorithms, REST API design, and System Design fundamentals — including scalability, caching, load balancing, database sharding/replication, and microservices architecture. Shipped production systems serving 200+ active users and co-founded a startup selected for the IIT Madras Nirmaan Stage 1 cohort. Pursuing BS in Electronic Systems at IIT Madras (GPA 8.32/10).

TECHNICAL SKILLS

- **Languages:** C/C++, Embedded C, Python, JavaScript, TypeScript, SQL, Linux Shell
- **Frontend:** React, Next.js (App Router), HTML, CSS, Tailwind CSS, shadcn/ui
- **Backend & Databases:** Node.js, Express, Fastify, REST APIs, MongoDB, PostgreSQL (Neon), Prisma ORM
- **System Design:** High-Level & Low-Level Design, Scalability & Load Balancing, Caching (Redis/CDN), SQL vs NoSQL & CAP Theorem, Indexing/Sharding/Replication, Consistency Models, Message Queues & Pub/Sub (Kafka/RabbitMQ), Microservices vs Monolith, Rate Limiting, Distributed Systems Fundamentals (Paxos/Raft, Leader Election, Quorum Reads/Writes)
- **Auth & Security:** JWT, RBAC, Clerk, OAuth basics, Secure API Design
- **Embedded / Hardware:** STM32 HAL, FreeRTOS, PWM, Raspberry Pi 5, nRF52840, PCB Design, MATLAB, LTSpice
- **DevOps / Tools:** Git & GitHub, Vercel, CI/CD basics, Docker (familiar), Postman, Cloudinary, VS Code
- **Core CS:** Data Structures & Algorithms, OOP, System Design Fundamentals, Distributed Systems Basics, Performance Optimization

EXPERIENCE

Co-founder & Full-Stack Developer — SelfShft *June 2026 – Present*

Selected, IIT Madras Nirmaan Stage 1 Cohort

- Built a full-stack web platform using Next.js, TypeScript, and PostgreSQL to connect students with location-based part-time jobs and short-term gigs.
- Designed and implemented core features for student onboarding, employer profiles, job matching, and authentication.
- Collaborated on product strategy and technical architecture as part of the founding team, addressing financial accessibility for relocating students.

Embedded Systems / Firmware Intern — BluArmor (Instrumentation Team) *Aug 2025 – Oct 2025*

Pre-Placement Offer (PPO) extended upon completion

- Designed automated RF/Bluetooth characterization systems using STM32 firmware, relay matrices, and ADC acquisition, reducing manual testing time by 15%.
- Built Python-based calibration and automation workflows, improving production measurement accuracy by 33%.
- Developed Python GUI tools for Bluetooth RSSI testing with automated CSV logging and live visualization, increasing throughput to 500 devices/day.
- Improved RSSI measurement stability by 55% using RF shielding and Exponential Moving Average (EMA) filtering.
- Automated headset pairing and validation workflows on Raspberry Pi 5 and nRF52840, enabling parallel multi-device testing.

PROJECTS

AI-Powered SaaS Platform *Next.js · Prisma · NeonDB · Clerk · Cloudinary*

- Architected a scalable multi-tenant SaaS system supporting 50+ concurrent tenants with role-based authentication.
- Integrated Cloudinary for optimized media uploads and transformations, reducing media load time by ~40%.
- Deployed serverless AI workloads with high-availability auto-scaling on Vercel edge infrastructure.

Full-Stack Auth & Admin Notes Platform *Node.js · MongoDB · Express · JWT · RBAC*

- Built a JWT + RBAC authentication system deployed to 200+ active IIT Madras users.
- Designed indexed MongoDB schemas that reduced query time by ~30%; implemented secure note approval workflows.

VibeCode Editor *React · Zustand · Appwrite · TypeScript*

- Developed a high-performance browser-based IDE with a virtualized file explorer and live execution playground, achieving sub-50ms re-render latency.
- Managed complex application state with Zustand and used Appwrite BaaS for secure schemas and real-time sync.

Fastify Backend & Thumbnailing Service *Fastify · Node.js Streams · JWT · PostgreSQL*

- Built a scalable backend using Fastify with modular architecture and Node.js streams for efficient file handling up to 100MB.
- Implemented JWT authentication and structured logging; achieved average API response times under 80ms.

EDUCATION

BS, Electronic Systems — Indian Institute of Technology Madras *2023 – 2027*

GPA: 8.32 / 10

System Design Mastery Series — Shivansh Vasu *Certificate*

30-day intensive covering scalability, load balancing, caching, database sharding & replication, consistency models, message queues, microservices, distributed systems fundamentals (Paxos/Raft), and FAANG-style case studies (Uber, YouTube/Netflix, Instagram).

SCHOLASTIC ACHIEVEMENTS

- JEE Main Percentile: 99.1
- LeetCode: Solved 300+ problems across core DSA patterns (sliding window, graphs, DP, greedy); active contest participant.
- Gold Medalist, Maths Olympiad (1st Rank, School Level).
- Science Olympiad (SOF): Secured All-India Rank 62, 2019.
- International Reasoning & Aptitude Olympiad (SilverZone): Achieved Rank 96, 2018.

LEADERSHIP & POSITIONS OF RESPONSIBILITY

- Deputy Head, Anukriti — Paradox, IIT Madras (Mar 2025).
- Teaching Assistant, Embedded C Programming Lab & Electronic Systems Thinking Lab, IIT Madras (May – Sept 2025) — mentored 200+ students in embedded C, microcontroller debugging, and low-level memory optimization; guided circuit design and system-level troubleshooting.
- Coordinator, Spark (ES Lab Activities), IIT Madras (Jan 2025).
- Group Leader, Kanha House, IIT Madras (Jan – Aug 2024); Logistics Team Coordinator (Oct 2023 – Jul 2024).